

Solution Requirements

Date	07 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely access the Smart Lender application (if login is implemented).	Medium
2	Authorization levels	Bank staff can access the prediction system and evaluate loan applications.	Medium

3	External Interfaces	The application reads the loan dataset and loads the trained XGBoost model for prediction.	High
4	Transactions processing	The system accepts applicant details, processes them through the ML model, and returns the loan approval result instantly.	High
5	Reporting	Display the loan prediction result (Approved/Rejected) to the user.	High
6	Business rules, etc.	Loan approval depends on applicant information such as income, loan amount, credit history, employment status, and property area.	High
7	Compliance to laws or regulations	Ensure applicant data is handled securely and only used for loan prediction purposes.	Medium
8	<i>Other</i>	Compare multiple ML algorithms and use the best-performing model (XGBoost) for deployment.	High

Step 2: Non-Functional Requirements (NFR)

1	Performance & Speed	Generate loan prediction quickly.	Prediction within 2 seconds
2	Scalability	Support multiple users submitting loan applications simultaneously.	Support 100+ concurrent users
3	Security & Data Privacy	Protect applicant information and prevent unauthorized access.	Secure handling of user input and model files
4	Reliability & Availability	The application should provide consistent predictions with minimal downtime.	99% availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy to use.	User-friendly interface accessible on modern browsers
6	<i>Other</i>	The application should be maintainable and easy to update with new models or datasets.	Modular and well-documented code